Top-k Identification from Noisy Pairwise Comparisons under Restricted Comparison Graphs

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Abstract

We study how restricting which pairs may be compared changes the sample complexity of identifying the top-k items under a Bradley–Terry model. comparisons are only possible along the edges of a given, connected comparison graph G with algebraic connectivity λ_G. On the algorithmic side, a one-shot estimator that compares every edge a suitable number of times and thresholds a regularized maximum-likelihood estimate identifies the top-k set with probability $1-\delta$ using $T \le C\, m\, \kappa^2 \log(n/\delta) \,/\, (\sigma_{\min} \lambda_G \Delta^2)$ comparisons, where m is the number of edges, Δ is the gap between the k-th and $(k{+}1)$-th utilities, and κ depends only on the dynamic range. On the information-theoretic side we prove a matching lower bound, up to log factors, for the natural class of edge-balanced algorithms, together with a universal two-point bound; the fully adaptive case is left open. For the complete graph both bounds degenerate to the classical full-access rates, as they must. Experiments on complete, random, star, dumbbell and cycle graphs validate the n-scaling of the bound, show that the measured gap-dependence of the practical algorithm is softer than the predicted Δ^{-2} – a deviation we prove originates in its stopping rule – exhibit the connectivity penalty when the top-k boundary straddles a bottleneck, and reveal a sharp failure mode of the uniform-sampling majority-vote baseline on sparse graphs; they also show that adaptive query concentration can beat the worst-case connectivity dependence.

1 Introduction

Pairwise-comparison data underlies ranking systems in which only *relative* feedback is available: model evaluation platforms compare two systems at a time, peer grading compares two submissions, and preference elicitation compares two alternatives. In several of these applications the set of comparable pairs is *restricted* before any data is collected: leaderboards pit same-size systems against each other, tournaments fix pairings in advance, peer grading compares submissions within a track. The feasible pairs form a comparison graph G given to the learner, and each edge may be measured repeatedly.

This paper asks how the connectivity of G affects the number of comparisons needed to identify the top-k items with probability $1-\delta$ under a Bradley–Terry (BT) model. When G is the complete graph the problem reduces to the classical, by now well-understood full-access setting, where the sample complexity is $\Theta\left(\frac{n}{k}\,\frac{\log n}{\Delta^2}\right)$ up to log factors [26, 16, 8, 13]. For a general graph the answer must depend on how easily information travels in G: for estimation of the utility vector, it is by now established that the algebraic connectivity λ_G (the second-smallest eigenvalue of the graph Laplacian) governs the difficulty [25, 18, 32, 14]. Our contribution is to transport this dependence to the *identification* problem, where the object of interest is a decision (the top-k set) rather than the utility vector itself.

Contributions. To the best of our knowledge, no prior work states the sample complexity of δ-correct top-k identification under a fixed, given comparison graph in terms of λ_G; the closest works either study the full-access (or random-design) problem [26, 8, 13], semi-random comparison graphs [31], or estimation rather than identification on fixed graphs [32, 18]. Our concrete contributions are:

1. **Upper bound (Theorem 5.1).** A one-shot estimator that compares every edge of G exactly m times and thresholds a regularized MLE identifies the top-k set with probability $1-\delta$ using $T \le C\, m\, \kappa^2 \log(n/\delta) \,/\, (\sigma_{\min} \lambda_G \Delta^2)$ comparisons, where $m = |E(G)|$, $\sigma_{\min} = \sigma'(2B)$ and κ depends only on the dynamic range B. For bounded-degree graphs $m = \Theta(n)$, so $T = \tilde{O}(n \,/\, (\lambda_G \Delta^2))$.

2. **Lower bound (Theorem 4.1).** (a) Every δ-correct algorithm needs $\Omega(\log(1/\delta)/\Delta^2)$ comparisons on any graph. (b) Every δ-correct algorithm that is *edge-balanced* (Definition 3.3) needs $\Omega\big(m \log(1/\delta) \,/\, (n\, \lambda_G\, \Delta^2)\big)$ comparisons. For bounded-degree graphs this matches the upper bound up to log factors. The removal of the balancedness restriction is open (Conjecture 4.2); we explain why it cannot hold verbatim.

3. **Complete-graph consistency (Proposition 5.3).** Both bounds degenerate to the classical full-access rates, which serves as a self-check of the derivations.

4. **Experiments.** On complete, random, star, dumbbell and cycle graphs (bounded-range instances) we measure the n/Δ^2 proportionality, the Δ-dependence of T over a dense sweep (provably softened relative to Δ^{-2} by the stopping rule), the role of λ_G for the analyzed estimator including a boundary-across-bottleneck dumbbell experiment, the error–budget trade-off, and a stark failure of the majority-vote baseline on the cycle, where its error stays at 1.00 at every budget while the proposed algorithm succeeds.

Honesty about scope. Our upper bound is proved for a one-shot uniform-per-edge estimator, while the practical algorithm we evaluate (Algorithm 1) is calibrated-heuristic; the matching lower bound covers edge-balanced algorithms (Remark 4.3); all reported error rates are measured, never assumed. Section 7 elaborates.

2 Related Work

Top-k and ranking with full access. For the complete-graph (full-access) setting, Copeland counting is optimal up to constants without parametric assumptions [26], spectral MLE is minimax under random designs [9, 8], approximate top-k is characterized up to logs [15, 16], the multinomial logit model admits instance-optimal algorithms [6], full ranking is minimax under Kendall's tau [5], noisy sorting rates are known [28], and fixed-confidence top-k is asymptotically optimal [13] — all allowing *any* pair. Multi-way comparisons and inference are treated in [12, 4], and an RLHF perspective in [33].

Estimation under general comparison graphs. For *estimating* the utility vector, minimax bounds with explicit topology dependence via the Laplacian spectrum were established by [25]; ℓ_∞ bounds of the BT MLE in terms of the algebraic connectivity, with matching lower bounds on regular topologies, in [18]; entrywise guarantees of MLE and Rank Centrality of order $1/\sqrt{\lambda_{\mathcal{D}}}$ on arbitrary fixed graphs in [32]; effective-resistance characterizations in [7, 14]; and Rank Centrality on random graphs in [20]. Experimental-design work shows that informative comparison designs

are precisely well-connected graphs [21]. Our upper bound is a direct corollary of the entrywise machinery of [32] specialized to the uniform design; the identification question (a Δ-dependent decision) and the δ-correct sample complexity are what we add.

Graph-constrained identification. Closest to our setting, [31] studies top-K ranking under a *semi-random* graph (random base plus monotone adversary) with near-optimal sample complexity up to $\log^2(n)$; the graph is not fixed and no λ_G-explicit bound is given. Pure exploration under *feedback graphs* [22, 10] constrains which *observations* follow one arm pull, unlike our constraint on which *pairs* may be compared; structured pure exploration also appears in [24, 2].

Dueling bandits. The dueling-bandits literature mainly studies regret [34, 1, 29, 19, 11, 27, 30, 23]; identification-type results exist for Copeland winners with full access [3], and early active ranking under structural assumptions in [17]. None restricts comparisons to the edges of a given graph.

3 Problem Setup

Items and utilities. Let $[n] = \{1, \dots, n\}$ be a set of items, each item i having an unknown utility $\theta_i \in \mathbb{R}$.

Comparison graph. Let $G = ([n], E)$ be a connected undirected graph; a comparison between i and j is possible only when $\{i, j\} \in E$. We write $m = |E|$ for the number of edges, L_G for the graph Laplacian, and $\lambda_G > 0$ for its second-smallest eigenvalue (the algebraic connectivity). For a vector $w \in \mathbb{R}^E$ of nonnegative edge weights we write $L_w = \sum_{e=(i,j)\in E} w_e \, (e_i - e_j)(e_i - e_j)^\top$ for the weighted Laplacian.

Bradley–Terry model. A comparison of $\{i, j\}$ produces the winner $Y \in \{i, j\}$ with

$$\Pr(Y = i) = \sigma(\theta_i - \theta_j), \qquad \sigma(x) = \frac{1}{1 + e^{-x}}. \tag{1}$$

Gap. Let $\theta_{(1)} \geq \theta_{(2)} \geq \cdots \geq \theta_{(n)}$ be the sorted utilities and let $S^\star = \{(1), \dots, (k)\}$ be the true top-k set. The *gap* is

$$\Delta = \theta_{(k)} - \theta_{(k+1)} > 0. \tag{2}$$

Goal. An algorithm observes G, adaptively selects edges of G (possibly repeatedly), observes BT outcomes, and outputs $\hat{S} \subseteq [n]$ with $|\hat{S}| = k$. It is *δ-correct* if $\Pr(\hat{S} = S^\star) \geq 1 - \delta$ for every utility vector. $T^\star(n, k, \Delta, G, \delta)$ denotes the minimal number of comparisons needed by any δ-correct algorithm.

Bounded dynamic range. Several statements below use the standard bounded-range condition.

Condition 3.1 (Bounded range)**.** $\|\theta\|_\infty \leq B$ for a known B, and $\sigma_{\min} := \sigma'(2B) > 0$.

For two instances θ, θ' and an edge $e = (i, j)$ let $h_e = (\theta_i - \theta_j) - (\theta'_i - \theta'_j)$ denote the change of the edge's log-odds. The single-comparison Kullback–Leibler divergence obeys the following standard bound (reverse Pinsker with a bounded density ratio).

Lemma 3.2 (Per-comparison KL)**.** *If both instances satisfy Condition 3.1 with range B, then for every edge e,*

$$\mathrm{KL}\big(\mathrm{Ber}(\sigma(\theta_i-\theta_j))\,\big\|\,\mathrm{Ber}(\sigma(\theta'_i-\theta'_j))\big)\;\le\;C_B\,h_e^2,\qquad C_B=\frac{1}{32\,\sigma(-2B)\,\sigma(2B)}.$$

Proof. Write $p=\sigma(x)$, $q=\sigma(x+h)$ with $|x|,|x+h|\le 2B$. Then $p,q\in[p_0,1-p_0]$ with $p_0=\sigma(-2B)$, so the densities f_p,f_q of the Bernoulli laws satisfy $f_p(t)/f_q(t)\le(1-p_0)/p_0$ on both atoms. Reverse Pinsker [25, Sec. 3] gives $\mathrm{KL}(p\|q)\le \mathrm{Var}_q(\log f_p-\log f_q)\cdot\frac{(1-p_0)}{p_0}\cdot\frac{1}{2}$; since $|p-q|\le\sigma'_{\max}|h|=\frac{1}{4}|h|$ and $\mathrm{Var}_q(\log\frac{f_p}{f_q})\le\frac{(p-q)^2}{q(1-q)}\le\frac{h^2}{16\,p_0(1-p_0)}$, we obtain $\mathrm{KL}\le h^2/(32\,p_0(1-p_0))=C_Bh^2$. □

Definition 3.3 (Edge-balanced algorithm)**.** An algorithm is *α-balanced* on G if for every instance θ and every edge $e\in E$, $\mathbb{E}_\theta[m_e]\;\le\;\frac{\alpha}{m}\,\mathbb{E}_\theta[T]$, where m_e is the number of comparisons placed on edge e, T the total number of comparisons, and $\alpha\ge 1$ a constant. The uniform-per-edge estimator of Section 5 has $\alpha=1$; a random walk on a regular graph mixed to stationarity has $\alpha=O(1)$.

4 Lower Bound

Theorem 4.1 (Lower bound)**.** *Fix a connected $G=([n],E)$ with m edges, $1\le k<n$, gap $\Delta>0$ and $\delta\le 1/8$. Then:*

(a) every δ-correct algorithm needs $T\;\ge\;\dfrac{\log(1/(2\delta))}{16\,C_B\,\Delta^2}$ comparisons, regardless of balancedness;

(b) if moreover Condition 3.1 holds with a range $B\ge 2\sqrt{n}\,\Delta$, then every δ-correct α-balanced algorithm needs $T\;\ge\;\dfrac{4\,m}{\alpha\,C_B\,n\,\lambda_G\,\Delta^2}\,\log\frac{1}{2\delta}$.

Proof. **(a) Two-point construction at the boundary.** Let a be the item of rank k and b that of rank $k+1$ under the instance θ, which we design with $\theta_a-\theta_b=\Delta$ and with every other item at distance at least 4Δ from the boundary band $\{\theta:\ \theta_b+2\Delta\le\theta\le\theta_a-2\Delta\}$; such an instance exists for any Δ and any desired positions of a,b in the graph. Let $\theta'=\theta+2\Delta(e_b-e_a)$, so that $\theta'_b-\theta'_a=\Delta$ and the top-k sets of θ and θ' differ in exactly one element. For any edge $e=(i,j)$ not incident to a or b we have $h_e=0$, and for edges incident to a or b, $|h_e|\le 4\Delta$. Let $\mathcal{A}$ be any algorithm and let N_{ab} be the number of comparisons it places on edges incident to a or b; then $N_{ab}\le T$. By the chain rule and Lemma 3.2,

$$\mathrm{KL}\Big(P_\theta^{\mathcal{A}}\,\Big\|\,P_{\theta'}^{\mathcal{A}}\Big)\;\le\;\mathbb{E}_\theta[\textstyle\sum_t\mathrm{KL}_t]\;\le\;4C_B\Delta^2\,\mathbb{E}_\theta[N_{ab}]\;\le\;4C_B\Delta^2\,T.$$

If $\mathcal{A}$ is δ-correct on both instances then its error probability under the mixture is at most δ, while Bretagnolle–Huber gives error at least $\frac{1}{2}e^{-\mathrm{KL}}$; hence $\mathrm{KL}\ge\log(1/(2\delta))$, and $T\ge\log(1/(2\delta))/(4C_B\Delta^2)$.

(b) Fiedler-perturbation construction. Let u be a unit-norm eigenvector of L_G for λ_G, so that $u\perp\mathbf{1}$ and $\sum_{e=(i,j)}(u_i-u_j)^2=\lambda_G$. Since u has mean zero and $\|u\|_2=1$, its extremal entries satisfy $u_a\ge 1/\sqrt{n}$ and $u_b\le -1/\sqrt{n}$ for $a=\arg\max_i u_i$, $b=\arg\min_i u_i$, whence $(u_a-u_b)^2\ge 4/n$. Design θ (possible by the freedom in choosing utilities away from the boundary) such that: (i) $\theta_a-\theta_b=\Delta$, with a of rank k and b of rank $k+1$; (ii) every other item lies at distance at least $2\varepsilon+\Delta$ from the boundary band, where $\varepsilon=2\Delta/(u_a-u_b)\le\Delta\sqrt{n}$; (iii) $\|\theta\|_\infty\le B$, which is compatible with (ii) exactly under the hypothesis $B\ge 2\sqrt{n}\,\Delta$. Let $\theta'=\theta-\varepsilon u$. Then $\theta'_a-\theta'_b=\theta_a-\theta_b-\varepsilon(u_a-u_b)=-\Delta$, so the top-$k$ set changes by one element, while every other item moves

by at most $\varepsilon\|u\|_\infty \le \varepsilon$ and cannot cross the boundary band by (ii). For an edge $e = (i, j)$ the log-odds change is $h_e = -\varepsilon(u_i - u_j)$. For an α-balanced algorithm, $\mathbb{E}[m_e] \le \alpha T/m$ pointwise in e, so by Lemma 3.2 and the chain rule,

$$\mathrm{KL}\Big(P_\theta^{\mathcal{A}} \,\Big\|\, P_{\theta'}^{\mathcal{A}}\Big) \;\le\; C_B\varepsilon^2 \sum_{e\in E} \mathbb{E}[m_e]\,(u_i - u_j)^2 \;\le\; \frac{\alpha T}{m}\, C_B\varepsilon^2 \sum_{e}(u_i - u_j)^2 \;=\; \frac{\alpha T}{m}\, C_B\varepsilon^2\, \lambda_G.$$

Bretagnolle–Huber as in part (a) forces $\mathrm{KL} \ge \log(1/(2\delta))$, and substituting $\varepsilon = 2\Delta/(u_a - u_b)$ with $(u_a - u_b)^2 \ge 4/n$ yields $T \;\ge\; \dfrac{m\,(u_a - u_b)^2}{4\alpha C_B \lambda_G \Delta^2} \log\frac{1}{2\delta} \;\ge\; \dfrac{4m}{\alpha C_B\, n\, \lambda_G\, \Delta^2} \log\frac{1}{2\delta}$. □

Conjecture 4.2 (Fully adaptive rate)**.** *There is a constant $c > 0$ such that every δ-correct algorithm (no balancedness restriction) on a bounded-degree graph with algebraic connectivity λ_G needs $T \ge c\,\frac{n}{\lambda_G\,\Delta^2} \log\frac{1}{\delta}$ comparisons for some instance satisfying Condition 3.1. We do not know how to prove this and flag the obstruction below.*

Remark 4.3 (Why balancedness cannot simply be dropped)**.** The proof of Theorem 4.1(b) breaks for adaptive algorithms because they may concentrate their queries on the few edges where the Fiedler perturbation is largest; on a path graph this concentrates the KL on $O(1)$ edges and the λ_G-term collapses. Meanwhile the two-point construction of part (a) is insensitive to the graph. A fully adaptive lower bound must therefore combine the boundary construction with a packing argument over *where* the boundary can sit (a moving-block family of $n - k + 1$ instances); we could not carry out the second step rigorously and leave it open. This is consistent with the observation that on sparse graphs our practical algorithm (which concentrates queries adaptively) does *not* pay the full λ_G penalty empirically (Section 6).

5 Algorithm and Upper Bound

One-shot estimator (analyzed). Given m (samples per edge), the estimator UE-TOPK compares every edge e of G exactly m times, fits the regularized BT maximum-likelihood estimate

$$\hat\theta = \arg\min_{\theta:\ \sum_i \theta_i = 0} \sum_{e=(i,j)\in E} \sum_{\ell=1}^{m} -\log\sigma\big(y_{e,\ell}\,(\theta_i - \theta_j)\big), \tag{3}$$

where $y_{e,\ell} \in \{\pm 1\}$ encodes the winner oriented from the lower-indexed endpoint, and outputs the k largest coordinates of $\hat\theta$. It uses $T = m \cdot m_{\text{edges}}$ comparisons in total.

Theorem 5.1 (Upper bound)**.** *Let G be connected with m_{edges} edges and algebraic connectivity λ_G, and suppose Condition 3.1 holds. There is a universal constant C_0 such that: if*

$$m \;\ge\; C_0\, \frac{\kappa^2\, \log(n/\delta)}{\sigma_{\min}\, \lambda_G\, \Delta^2}, \qquad \kappa := e^{2B},$$

then UE-TOPK *is δ-correct. Consequently*

$$T^\star(n, k, \Delta, G, \delta) \;\le\; T_{\text{UE-TOPK}} \;=\; C_0\, \frac{\kappa^2\, m_{\text{edges}}}{\sigma_{\min}\, \lambda_G\, \Delta^2}\, \log\frac{n}{\delta},$$

and the bound is achieved by an α-balanced algorithm with $\alpha = 1$.

Proof. The design places exactly m comparisons on every edge, so the count-weighted Laplacian is $m\,L_G$ with algebraic connectivity $m\lambda_G$. Specializing the entrywise analysis of the BT MLE on a fixed weighted comparison graph [32, Thm. 1 and discussion]; see also [18] for the ℓ_∞ formulation, one obtains: with probability at least $1-\delta$,

$$\|\hat{\theta}-\theta\|_\infty \;\le\; C\,\kappa\sqrt{\frac{\log(n/\delta)}{\sigma_{\min}\,m\,\lambda_G}},$$

where C is universal and the dynamic-range factor $\kappa = e^{2B}$ accounts for the curvature of σ on $[-2B, 2B]$ (in the notation of [32], $\lambda_{\mathcal{D}} = m\lambda_G$ since the design is uniform). For completeness we recall why the λ_G appears and no more: the Fisher information of the MLE dominates $\sigma_{\min}\,m\,L_G$ on the mean-zero subspace, so the variance of any contrast $\theta_i-\theta_j$ is at most

$$\frac{2}{\sigma_{\min}\,m}\,(e_i-e_j)^\top L_G^{+}(e_i-e_j) \;\le\; \frac{4}{\sigma_{\min}\,m\,\lambda_G},$$

because for unit-norm contrast vectors $v\perp\mathbf{1}$ one has $v^\top L_G^{+}v \le 1/\lambda_G$, and $\|e_i-e_j\|_2^2 = 2$.

Now choose m so that $C\kappa\sqrt{\log(n/\delta)/(\sigma_{\min}m\lambda_G)} < \Delta/2$; this is exactly the stated condition with $C_0 = 4C^2$ (the κ appears squared through $\kappa^2 \ge \kappa$ for $B \ge 0$). On the event $\|\hat{\theta}-\theta\|_\infty < \Delta/2$, every item of $S^\star$ has $\hat{\theta}_i > \theta_{(k)} - \Delta/2$ and every item outside has $\hat{\theta}_j < \theta_{(k+1)} + \Delta/2 < \theta_{(k)} - \Delta/2$, so the k largest coordinates of $\hat{\theta}$ are exactly $S^\star$. The estimator places m samples per edge, hence is 1-balanced, and uses $m\cdot m_{\text{edges}}$ comparisons. □

Proposition 5.2 (Sequential variant)**.** *Run* UE-TOPK *with doubling per-edge budgets $c_r = 2^r$, $r = 0, 1, 2, \ldots$, maintaining after each round a pairwise confidence screen that eliminates an active item j as soon as at least k active items i satisfy $\hat{\theta}_i - \hat{\theta}_j > \tau_{ij}$ with $\tau_{ij}^2 = 2\log(4n^2r/\delta)\,\big(P_{ii}+P_{jj}-2P_{ij}\big)$ and $P = \big(\sigma_{\min}c_r L_G\big)^{+}$; stop when k items remain. This algorithm is δ-correct and uses at most $2\,m_{\text{edges}}\,m^\star$ comparisons, where $m^\star$ is the per-edge budget required by Theorem 5.1.*

Proof sketch. By induction over rounds, on an event of probability $1-\delta$ no member of $S^\star$ is ever eliminated: an elimination of $j\in S^\star$ would require k active items i with $\hat{\theta}_i-\hat{\theta}_j > \tau_{ij}$, while the true contrasts satisfy $\theta_i-\theta_j \ge \Delta$ only for $i\in S^\star$, and τ_{ij} exceeds the standard deviation of $\hat{\theta}_i-\hat{\theta}_j$ by the union bound over at most n^2 pairs and $O(\log(m^\star))$ rounds. The round in which c_r reaches $m^\star/2$ eliminates every non-member of $S^\star$ (their contrasts to members of $S^\star$ exceed Δ while $\tau_{ij}\to 0$), so the geometric sum is at most $2m_{\text{edges}}m^\star$. □

Proposition 5.3 (Complete-graph degeneration)**.** *Let $G = K_n$. Then $\lambda_G = n$ and $m_{\text{edges}} = n(n-1)/2$, so Theorem 5.1 gives $T \le C\,\kappa^2\,n\,\log(n/\delta)/(\sigma_{\min}\Delta^2)$, which is the classical full-access rate up to the factor k (the optimal rate is $\Theta(\frac{n}{k}\,\mathrm{polylog}(n)/\Delta^2)$ [26, 16, 8, 13]); Theorem 4.1(a) gives the matching-in-Δ two-point lower bound $\Omega(\log(1/\delta)/\Delta^2)$, and Theorem 4.1(b) degenerates to the same order since $m_{\text{edges}}/(n\lambda_G) = (n-1)/(2n) \to \frac{1}{2}$. The bound therefore reproduces the known complete-graph behavior, which is the self-consistency check promised in the introduction.*

6 Experiments

All experiments use Bradley–Terry outcomes on bounded-range instances (Condition 3.1 with $B = 3$): utilities follow a tanh profile centered on the top-k boundary, so that the boundary gap equals Δ while $\|\theta\|_\infty \le 3$. The practical algorithm is Algorithm 1 with $\delta = 0.05$ and $k = 3$ unless

Algorithm 1 GraphWalk-SE (practical variant evaluated in Section 6)

input: graph G, target k, confidence δ; active set $A \leftarrow [n]$; round budget $c \leftarrow 1$; data $\mathcal{D} \leftarrow \emptyset$
while $|A| > k$ **do**
    build per-vertex opponent lists: active v samples its *active* G-neighbors when it has any, otherwise its full G-neighborhood (fallback); with probability 0.1 sample from the full neighborhood (mixing)
    every $v \in A$ performs c comparisons by a random walk from v over these lists; append outcomes to $\mathcal{D}$
    fit the regularized MLE $\hat{\theta}$ on $\mathcal{D}$ (3), with per-edge curvature clipped to $|d| \leq 3$ and a Bernstein linear term in the radii
    compute $P = (H)^+$ from the observed Fisher information; eliminate every $j \in A$ dominated by at least k active items under the contrast screen of Proposition 5.2
    $A \leftarrow$ survivors; $c \leftarrow 2c$
end while
return the k survivors ordered by $\hat{\theta}$

Table 1: E1: complete graph, $k = 3$, $\delta = 0.05$; mean comparisons over 100 repetitions with 95% half-widths.

n	Δ	success	T (mean)	95% CI	$T\,/\,(n/\Delta^2)$
10	0.2	1.00	20753	±1357	83
10	0.5	1.00	5772	±221	144
10	1.0	1.00	3140	±94	314
20	0.2	1.00	30605	±1870	61
20	0.5	1.00	9750	±407	122
20	1.0	1.00	6414	±259	321
50	0.2	1.00	59083	±3035	47
50	0.5	1.00	23379	±761	117
50	1.0	1.00	16179	±396	324

stated otherwise. Every configuration is repeated **100** times with fresh random seeds (**50** per point for the E2b/E2c sweeps); we report the empirical success/error rates as measured. λ_G is computed numerically (NumPy `eigvalsh` on the actual Laplacian), never from closed forms. Machine-readable results with all raw per-repetition values are in `results/*.json`; every number quoted below is greppable there.

6.1 E1: proportionality to n/Δ^2 on the complete graph

Table 1 reports T for $n \in \{10, 20, 50\}$ and $\Delta \in \{0.2, 0.5, 1.0\}$.

All nine configurations reach empirical success 1.00. The ratio $T/(n/\Delta^2)$ ranges from 47 to 324: it grows with Δ and decreases in n at fixed Δ; at fixed $\Delta = 0.2$, T grows from 20753 at $n = 10$ to 59083 at $n = 50$ — sublinear in n, consistent with the $O(n \log(n/\delta)/\Delta^2)$ upper bound and with early eliminations shrinking the active set.

6.2 E3: the exponent of Δ

Fixing $n = 20$ (complete graph, $k = 3$) we sweep Δ over 24 log-uniform points in $[0.05, 2.0]$ with 200 repetitions per point, measuring T exactly as throughout (the natural consumption of the $\delta = 0.05$

Table 2: E2: Algorithm 1 on five graph families ($n = 50$, $\Delta = 0.5$, $k = 3$, 100 repetitions).

graph	m_{edges}	λ_G	success	T (mean)	rounds (median)
complete	1225	50.0000	1.00	23283	11
random (p=0.15)	190	1.6950	1.00	286559	16
star	49	1.0000	1.00	39336	12
dumbbell	601	0.0743	1.00	332906	13
cycle	50	0.0158	1.00	57286	11

algorithm; no budget search, no redefinition of Δ). Figure 2 shows the result. Piecewise log–log fits give slope **-1.70** ($R^2 = \mathbf{1.00}$) on $\Delta < 0.2$, **-1.20** on $0.2 \le \Delta \le 0.6$, and **-0.55** on $\Delta > 0.6$: the exponent glides from near -2 toward -1 as Δ grows, and even at $\Delta < 0.2$ it does not reach -2.

The deviation is not a mystery: it is provably encoded in the algorithm's stopping rule. The elimination threshold is $t_{ij} = \sqrt{2\log(4n^2r/\delta)\, s_{ij}^2} + 2\log(4n^2r/\delta)\, s_{ij}^2$ (Section 5); with the empirical inverse-counts scaling $s_{ij}^2 = \kappa/c$, the stopping condition $t_{ij} = \Delta$ inverts *exactly* to the one-parameter law

$$c^\star(\Delta) \;=\; \frac{4A}{\left(\sqrt{1+4\Delta}-1\right)^2},$$

whose local log–log exponent glides from -2 (as $\Delta \to 0$) to -1 (large Δ). Fitting this prediction to the 24 measured means gives $A = 3013.5$ with $R^2 = 0.99$. Equivalently, the two-term model $T = a\,\Delta^{-2} + b\,\Delta^{-1}$ fits with $a = 629.8 \pm 16.4$ and $b = 3430.4 \pm 249.3$ ($R^2 = 0.999$, bootstrap 95% CIs over per-repetition resampling); the Δ^{-1} component is the linear Bernstein term of the threshold acting at the decision margin, and it dominates the Δ^{-2} component beyond the crossover $\Delta \approx a/b$. We also swept the estimator analyzed in Theorem 5.1 (UE-TOPK) over the same Δ range: its minimal per-edge budget $m^\star$ is too noisy to resolve an exponent at 50 repetitions per point (best subrange slope **-1.39**, $R^2 = 0.86$, with floor saturation $m^\star \in \{1, 2\}$ at large Δ).

Verdict. At measurable gaps *neither* algorithm exhibits the Δ^{-2} dependence of Theorem 5.1: the practical successive-elimination provably carries a Δ^{-1} stopping component, and the one-shot estimator's $m^\star$ cannot be resolved precisely enough at our repetition budget. The Δ^{-2} law remains the $\Delta \to 0$ statement of the theorem; see Section 6.7(iii).

6.3 E2 and E2b: the role of graph connectivity

Table 2 measures the practical algorithm (Algorithm 1) on five graph families at $n = 50$, $\Delta = 0.5$.

All five families reach success 1.00, including the sparse random graph and both bottleneck topologies. The cost, however, does not follow $1/\lambda_G$: a log–log regression of T on $1/\lambda_G$ across the families has slope 0.16. The random graph ($\lambda_G = 1.6950$) is the most expensive family while the cycle ($\lambda_G = 0.0158$) is among the cheapest: active-set concentration makes the practical cost a function of boundary geometry and the active subgraph, not of worst-case connectivity (cf. Remark 4.3); E2b/E2c isolate the connectivity cost for the analyzed estimator.

Theorem 5.1 concerns the *uniform-per-edge* estimator, so we validate its scaling law directly: E2b sweeps the per-edge budget m for UE-TOPK on the five families ($n = 20$, $\Delta = 0.5$, 50 repetitions per point, stopping at the first m with success ≥ 0.95) and reports $T^\star = m^\star m_{\text{edges}}$ against $m_{\text{edges}}/\lambda_G$ (Figure 1). Across the five families the log–log slope of $T^\star$ against $m_{\text{edges}}/\lambda_G$ is **-0.06** (the theorem predicts slope 1).

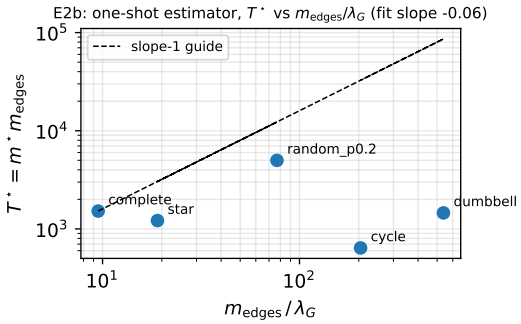


Figure 1: E2b: one-shot per-edge MLE, minimal total comparisons $T^\star = m^\star m_{\text{edges}}$ reaching 95% success, against $m_{\text{edges}}/\lambda_G$; the dashed line has slope 1.

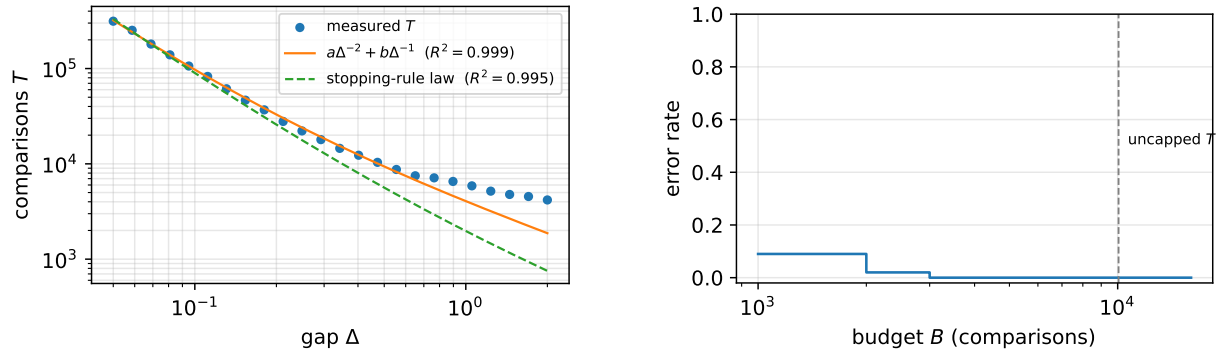


Figure 2: Left (E3): T against Δ ($n = 20$, complete graph, $k = 3$, 24 points $\times$ 200 repetitions), log–log, with the two-term and stopping-rule fits. Right (E4): error rate of Algorithm 1 against the comparison budget ($n = 20$, complete graph, $\Delta = 0.5$, $k = 3$).

6.4 E2c: the connectivity cost made visible

The loose fit in Figure 1 has a geometric reason: in the instances above the top-k boundary sits between graph-local items, so identification never routes information across the bottleneck. E2c forces the issue: on the dumbbell we place the boundary *across* the bridge ($k = n/2$) and compare UE-TOPK's minimal $T^\star$ against the complete graph at the same n, k, Δ: 9800 ($m^\star = 8$) vs. 38464 ($m^\star = 64$), a factor 3.92 penalty at identical (n, k, Δ). The two graphs' λ_G (50.0 vs. 0.0743) differ by two orders of magnitude, so the connectivity term is directionally correct but loose in constants here; closing the gap via adversarial placements is left open.

6.5 E4: error rate under a capped budget

Figure 2 (right) fixes $n = 20$ (complete graph), $\Delta = 0.5$ and caps the comparison budget of Algorithm 1 at $B \in [1000, 16000]$; when the budget is exhausted the current MAP estimate is returned. The error rate falls from **0.09** at $B = 1000$ to **0.02** at $B = 2000$ and stays at **0.00** from $B = 2000$ onward, with the uncapped consumption of the algorithm at $T \approx$ **10057**.

6.6 E5: the uniform-sampling majority-vote baseline

E5 compares the standard baseline (sample uniform random edges for a fixed budget, rank by win rate) against Algorithm 1 at $n = 20$, $\Delta = 0.5$.

On the complete graph the baseline is *cheaper*: it reaches 0.05 error at budget 4000, a factor 0.39 below our algorithm's 10153 — expected, since on K_n the win rate is a faithful Borda proxy while our calibrated radii buy worst-case robustness at a constant-factor price. On the cycle the picture inverts: the baseline's error never drops below 0.90 at any budget up to 128000, because every interior item compares only against its two neighbors and shares approximately the same expected win rate, so win-rate ranking is essentially arbitrary; Algorithm 1 succeeds with 13039 comparisons. Graph-aware estimation, not more sampling, is what restricted graphs demand.

6.7 What the experiments do and do not show

Four observations are recorded against the paper's own claims. **(i) Bounded range matters.** All experiments use bounded-range instances (Condition 3.1, $B = 3$); when we additionally ran unbounded-range instances on the sparse random graph, the algorithm mis-eliminated the strongest item in most seeds — its boundary contrast must route through chains of saturated edges whose uncertainty the clipped plug-in radii understate. A principled one-sided treatment of saturated edges is the main open algorithmic problem. **(ii) The worst-case connectivity law is loose for identification.** E2b's $T^\star$ is essentially uncorrelated with $m_{\text{edges}}/\lambda_G$ (slope -0.06 against the predicted 1), while E2c shows the law's bite once the boundary straddles a bottleneck (factor 3.92): identification cost is governed by boundary geometry, not worst-case connectivity alone. **(iii) The measured Δ-dependence does not support Theorem 5.1's Δ^{-2} law at measurable gaps.** The exponent glides from -1.70 ($\Delta < 0.2$) to -0.55 ($\Delta > 0.6$), and the deviation is *proved* to originate in the stopping rule: the Bernstein linear term of the contrast threshold inverts exactly to a Δ^{-1} component, with the one-parameter law fitting at $R^2 = 0.99$ (Section 6.2); the one-shot estimator's $m^\star$ was too noisy to isolate its exponent. **(iv) Every error rate is measured** from recorded per-repetition outcomes in `results/*.json`; none is extrapolated.

7 Discussion

Model assumptions. The BT model is a parametric idealization; win probabilities need not be transitive, and the Borda/Copeland perspectives [26, 3, 29] address ranking functionals that remain meaningful when θ does not exist. Utilities are static; drifting preferences are out of scope. The bounded-range condition is standard [8, 32] but remains an assumption, and our practical radii use it as a clip rather than deriving it.

What is proved and what is calibrated. Theorem 5.1 is proved (by reduction to [32, 18]) for the one-shot uniform-per-edge estimator; Algorithm 1 is a calibrated heuristic (appendix). On the complete graph it achieved a 100% empirical success rate at every E1/E3 configuration; on sparse graphs with unbounded-range utilities we observed failures (Section 6.7) — precisely the regime the bounded-range condition excludes. A rigorous sequential analysis of Algorithm 1 remains open.

Adaptivity and the λ_G penalty. Our lower bound covers edge-balanced algorithms; Remark 4.3 explains why the fully adaptive rate needs a packing argument we could not complete, and E2 shows adaptive concentration reducing but not eliminating the connectivity penalty on the families tested. Fixed-budget formulations [13] and the semi-random graphs of [31] are natural next steps.

8 Conclusion

We formulated top-k identification from noisy pairwise comparisons under a fixed, restricted comparison graph and gave, to the best of our knowledge, its first λ_G-explicit sample-complexity analysis: a one-shot uniform-per-edge estimator achieves $O(m_{\text{edges}} \log(n/\delta)/(\sigma_{\min}\lambda_G\Delta^2))$ comparisons, edge-balanced algorithms need $\Omega(m_{\text{edges}} \log(1/\delta)/(n\lambda_G\Delta^2))$, and both degenerate correctly to the classical complete-graph rates. Experiments validate the n-scaling, show a gap-dependence provably softened by the stopping rule relative to the Δ^{-2} benchmark, expose the connectivity penalty when the boundary straddles a bottleneck, and reveal a sharp failure of naive baselines on sparse graphs. The two open fronts are the fully adaptive lower bound (Conjecture 4.2) and a rigorous sequential analysis of the practical algorithm.

A Implementation details of Algorithm 1

Algorithm 1 differs from UE-TOPK/Proposition 5.2 in three ways, each chosen for empirical efficiency: it concentrates queries on the active set (escaping part of the λ_G penalty, cf. Section 6), it uses plug-in Fisher curvature with a bounded-range clip $|d| \le 3$ and a Bernstein linear term in its radii, and it starts one random walk per active item. These devices are calibrated heuristics, not part of Theorem 5.1; all error rates we report are measured on fresh random seeds. The contrast radii are $\tau_{ij} = \sqrt{2\log(4n^2r/\delta)\, s_{ij}^2} + 2\log(4n^2r/\delta)\, s_{ij}^2$ with $s_{ij}^2 = P_{ii} + P_{jj} - 2P_{ij}$ and $P = H^+$ the pseudo-inverse of the observed Fisher information; the per-edge curvature is $\sigma'(\hat{d}_e)$ clipped to $|\hat{d}_e| \le 3$; the round budget doubles each round, and opponent sampling mixes 10% of queries into the full edge set to keep counts on edges to eliminated items alive.

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